

Voiding Cystourethrogram (VCUG)

Patient Information · What to expect before your bladder and urethra X-ray

WARWIKI

A voiding cystourethrogram, or “VCUG,” is an X-ray test of your bladder and urethra. Pictures are taken while your bladder is filled with dye and while you urinate. It shows how your bladder fills and empties. It helps your team find problems such as urine flowing back toward the kidneys, a narrowing, or a leak.

About This Test

Your bladder stores urine. The urethra is the tube that carries it out of your body. A VCUG fills the bladder with a dye that shows up white on X-ray, then takes pictures while you urinate. Because it shows the bladder and urethra *at work*, it can show things a still picture cannot.

Your care team may order a VCUG to:

- See how your bladder fills and empties
- Check whether urine flows back toward the kidneys (reflux)
- Look at the bladder neck and urethra as you urinate
- Find a narrowing, a leak, or a small pouch (diverticulum)
- Check healing after bladder, prostate or urethral surgery

The test looks at the bladder and urethra. It is a picture that guides your care, not a treatment.

Is It Safe?

Yes. A VCUG is a common, low-risk test. It uses a small amount of X-ray, and the dye leaves your body in your urine. The most common problem is a mild urine infection from the catheter. That is why a clean method is used, and some teams give an antibiotic. Brief discomfort, mild burning or a little blood when you urinate afterward usually pass quickly.

Learn the Terms

Bladder

The organ that stores urine until you are ready to empty it.

Urethra

The tube that carries urine from the bladder out of the body.

Catheter

A thin, soft tube that drains or fills the bladder. The dye can go through one you already have, or one placed for the test.

Suprapubic tube (SPT)

A catheter that drains the bladder through a small opening in the lower belly. It can be used to fill the bladder for the test.

Contrast dye

A liquid that shows up white on X-ray.

Voiding

Urinating, or emptying your bladder. The key pictures are taken while you void.

Reflux

Urine flowing back from the bladder toward the kidneys.

Fluoroscopy

A type of X-ray that shows live, moving pictures.

Post-void image

A picture taken after you urinate, to see how well your bladder emptied.

Will it hurt? If you already have a catheter or suprapubic tube, the dye usually goes through it. Otherwise a thin catheter is placed first, with brief pressure or stinging. Feeling a full bladder is normal. The test takes about 30 minutes.

How to Get Ready

- If you have a **catheter or suprapubic tube**, leave it in. It is usually used to fill your bladder.
- You can **eat, drink** and take your **usual medicines**, unless your team tells you otherwise.
- You can usually **drive yourself home**.

Tell your team ahead of time if you:

- Have had a reaction to X-ray dye
- Take a blood thinner
- **Could be pregnant.** X-rays are usually avoided in pregnancy.
- Have signs of a urine infection, such as burning or fever. The test may be moved.

What Happens During the Test

- 1 You lie on an X-ray table.
- 2 **Dye goes into your bladder** through your suprapubic tube or catheter. If you have neither, a thin catheter is passed through the urethra.
- 3 As your bladder fills, you feel you need to urinate. Pictures are taken.
- 4 When full, you **urinate while the X-ray records**. This is the key part. If your team says it is safe, a urethral catheter is taken out first.
- 5 A last picture may show how well your bladder emptied.

After a urethral repair, your team may leave the catheter in and run the dye alongside it. If the X-ray shows a leak, the catheter usually stays longer and the test is repeated later.

After the Test

- You can usually go back to **normal activities right away**.
- **Mild burning** or a little **pink urine** for a day or so is common. Drink extra water.

Call the office that did your test if you have:

- A temperature of **100.4 °F (38 °C)** or higher, or chills
- Burning or pain that gets worse instead of better
- A catheter or tube that stops draining

Go to an emergency department if:

- You cannot urinate and your bladder feels full or painful
- Heavy bleeding, or urine like red wine or with clots
- You cannot reach the office and are worried

Three Things to Remember

1. A VCUG films your bladder and urethra with dye while you urinate.
2. Keep any catheter or tube you have. It is usually used for the test.
3. Mild burning or a little blood for a day is normal. Drink extra water.